Bracketing a root by bisection

Upper high school · analysis and algorithms · Synthesis problem · 12

Problem · Investigate, justify, conclude. For $f(x) = x^3 - x - 1$ on $[1,2]$, prove existence and uniqueness of a root α. Carry out three bisection steps. How many halvings guarantee that the final midpoint approximates α with error at most 0.001?